possible nearest neighbor held in the branch is no further than an epsilon factor from our current nearest neighbor.

Of course, there could be several points in the dataset that are within a factor $(1+\epsilon)$ from the true nearest neighbor, so we are not guaranteed that we get the second-closest point, nor the third etc...

However, the probability that these points exist are proportional to the size of the ring region with radii $nnDist$ and $nnDist/(1+\epsilon)$, so the smallest we set ϵ, the lowest the chances we are missing closer points.

Even more, a more accurate estimate of the probability is given by the area of the intersection of the aforementioned ring with the bounding envelope of the node we skip; figure 10.23 illustrates this idea, and shows the difference between Ss-trees and k-d trees and R-trees: the probability is maximum when the inner radius is just tangent to the area, and a sphere has a smaller intersection, with respect to any rectangle .